Algorithm Analysis

1. What Is Algorithm Analysis?

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3. Example: Anagram Detection

4. Benchmark of `Python` Data Structures: `Lists/Dictionaries`

2.1~2.2 What Is Algorithm Analysis?

An interesting question often arises. When two programs solve the same problem but look different, is one program better than the other?

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In order to answer this question, we need to remember that there is an important difference between a program and the underlying algorithm that the program is representing!

There may be many programs for the same algorithm, depending on the programmer and the programming language being used!

To explore this difference further, consider the function that computes the sum of the first integers. The algorithm uses the idea of an accumulator variable that is initialized to 0. The solution then iterates through the integers, adding each to the accumulator.

To explore this difference further, consider the function that computes the sum of the first n integers. The algorithm uses the idea of an accumulator variable that is initialized to 0. The solution then iterates through the n integers, adding each to the accumulator.

In [1]:
```
#include <iostream>
using namespace std;

long long sumOfN(long long n) {
    long long theSum = 0;
    for (long long i = 1; i <= n; i++) {
        theSum = theSum + i;
    }
    return theSum;
}

int main() {
    cout << sumOfN(10) << endl;
    return 0;
}
```

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Now look at the function below:

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In [2]:
```
#include <iostream>
using namespace std;

long long foo(long long tom) {
    long long fred = 0;
    for (long long bill = 1; bill <= tom; bill++) {
        long long barney = bill;
        fred = fred + barney;
    }
    return fred;
}

int main() {
    cout << foo(10) << endl;
    return 0;
}
```

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In [2]:
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#include <iostream>
using namespace std;

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    long long fred = 0;
    for (long long bill = 1; bill <= tom; bill++) {
        long long barney = bill;
        fred = fred + barney;
    }
    return fred;
}

int main() {
    cout << foo(10) << endl;
    return 0;
}
```

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At first glance it may look strange, but this function is essentially doing the same thing as the previous one. Here, we did not use good identifiers for readability, and we used an extra assignment statement that was not really necessary!

The function `sum_of_n()` is certainly better than the function `foo()` if you are concerned with readability. Easy to read and easy to understand is important for beginner. In this course, however, we are also interested in characterizing the algorithm itself.

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Algorithm analysis is concerned with comparing algorithms based upon the **amount of computing resources that each algorithm uses**.

The function `sum_of_n()` is certainly better than the function `foo()` if you are concerned with readability. Easy to read and easy to understand is important for beginner. In this course, however, we are also interested in characterizing the algorithm itself.

Algorithm analysis is concerned with comparing algorithms based upon the **amount of computing resources that each algorithm uses**.

We want to be able to consider two algorithms and say that one is better than the other because it is more efficient in its use of those resources or perhaps because it simply uses fewer.

There are two different ways to look at this. One way is to consider the **amount of space or memory** an algorithm requires to solve the problem.

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As an alternative to space requirements, we can analyze and compare algorithms based on the **amount of time** they require to execute.

One way is that we can measure the execution time for the function `sum_of_n()` to do a benchmark analysis. In `Python`, we can benchmark a function by noting the starting time and ending time within the system we are using.

In [3]:

```cpp
#include <iostream>
#include <chrono>
using namespace std;
using namespace std::chrono;

long long sumOfN2(long long n, double& seconds) {
    auto start = steady_clock::now();
    long long theSum = 0;
    for (long long i = 1; i <= n; i++) theSum = theSum + i;
    seconds = duration<double>(steady_clock::now() - start).count();
    return theSum;
}

int main() {
    double secs;
    cout << sumOfN2(10, secs) << endl;
    return 0;
}
```

```
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#include <iostream>
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    double secs;
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In [4]:
```cpp
#include <iostream>
#include <chrono>
using namespace std; using namespace std::chrono;

long long sumOfN2(long long n, double& seconds) {
    auto start = steady_clock::now();
    long long theSum = 0;
    for (long long i = 1; i <= n; i++) theSum = theSum + i;
```

```
    seconds = duration<double>(steady_clock::now() - start).count();
    return theSum;
}
int main() {
    for (int run = 0; run < 5; run++) {
        double secs;
        long long s = sumOfN2(100000, secs);
        printf("Sum is %lld required %10.7f seconds\n", s, secs);
    }
}
```

Sum is 5000050000 required 0.0001346 seconds

Sum is 5000050000 required 0.0001432 seconds

Sum is 5000050000 required 0.0001412 seconds

Sum is 5000050000 required 0.0001413 seconds

Sum is 5000050000 required 0.0001350 seconds

We discover that the time is fairly consistent. What if we run the function adding the first 1,000,000 integers?

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In [5]:
```
#include <iostream>
#include <chrono>
using namespace std; using namespace std::chrono;

long long sumOfN2(long long n, double& seconds) {
    auto start = steady_clock::now();
    long long theSum = 0;
    for (long long i = 1; i <= n; i++) theSum = theSum + i;
    seconds = duration<double>(steady_clock::now() - start).count();
    return theSum;
}
int main() {
    for (int run = 0; run < 5; run++) {
        double secs;
        long long s = sumOfN2(1000000, secs);
        printf("Sum is %lld required %10.7f seconds\n", s, secs);
    }
}
```

Sum is 500000500000 required 0.0014261 seconds

Sum is 500000500000 required 0.0014092 seconds

Sum is 500000500000 required 0.0015807 seconds

Sum is 500000500000 required 0.0012599 seconds

Sum is 500000500000 required 0.0014389 seconds

Now consider the following function, which shows a different means of solving the summation problem. This function takes advantage of a closed equation to compute the sum of the first integers without iterating.

Now consider the following function, which shows a different means of solving the summation problem. This function takes advantage of a closed equation $\sum_{i=1}^{n} i = \frac{(n)(n+1)}{2}$ to compute the sum of the first n integers without iterating.

In [6]:
```
#include <iostream>
#include <chrono>
using namespace std;
using namespace std::chrono;

long long sumOfN3(long long n, double& seconds) {
    auto start = steady_clock::now();
    long long theSum = (n * (n + 1)) / 2;
    seconds = duration<double>(steady_clock::now() - start).count();
    return theSum;
}

int main() {
    double secs;
    cout << sumOfN3(10, secs) << endl;
    return 0;
}
```

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If we do the same benchmark measurement for `sum_of_n_3()`, using four different values for (100,000, 1,000,000, 10,000,000, and 100,000,000), we get the following results:

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In [7]:
```
#include <iostream>
#include <chrono>
using namespace std; using namespace std::chrono;

long long sumOfN3(long long n, double& seconds) {
    auto start = steady_clock::now();
    long long theSum = (n * (n + 1)) / 2;
    seconds = duration<double>(steady_clock::now() - start).count();
    return theSum;
}
int main() {
    long long ns[] = {100000, 1000000, 10000000, 100000000};
    for (long long n : ns) {
        double secs;
        long long s = sumOfN3(n, secs);
        printf("Sum is %lld required %10.7f seconds\n", s, secs);
    }
}
```

Sum is 5000050000 required 0.0000001 seconds

Sum is 500000500000 required 0.0000000 seconds

Sum is 50000005000000 required 0.0000000 seconds

Sum is 5000000050000000 required 0.0000000 seconds

If we do the same benchmark measurement for `sum_of_n_3()`, using four different values for n (100,000, 1,000,000, 10,000,000, and 100,000,000), we get the following results:

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int main() {
    long long ns[] = {100000, 1000000, 10000000, 100000000};
    for (long long n : ns) {
        double secs;
        long long s = sumOfN3(n, secs);
        printf("Sum is %lld required %10.7f seconds\n", s, secs);
    }
}
```

Sum is 5000050000 required 0.0000001 seconds

Sum is 500000500000 required 0.0000000 seconds

Sum is 50000005000000 required 0.0000000 seconds

Sum is 5000000050000000 required 0.0000000 seconds

First, the times recorded above are shorter than any of the previous examples. Second, they are very consistent no matter what the value of n. It appears that `sum_of_n_3()` is hardly impacted by the number of integers being added.

Intuitively, we can see that the iterative solutions seem to be doing more work since some program steps are being repeated. Also, the time required for the iterative solution seems to increase as we increase the value of .

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However, if we ran the same function on a different computer or used a different programming language, we would likely get different results. It could take even longer to perform `sum_of_n_3()` if the computer were older.

We need a better way to characterize these algorithms with respect to execution time. The benchmark does not really provide us with a useful measurement because it is dependent on a **particular machine, program, time of day, compiler, and programming language.**

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We would like to have a characterization that is independent of the program or computer being used. This measure would then be useful for judging the algorithm alone and could be used to compare algorithms **across implementations**!

2.3 Big O Notation

If each of these steps is considered to be a **basic unit** of computation, then the execution time for an algorithm can be expressed as the number of steps required to solve the problem!

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Deciding on an appropriate basic unit of computation can be a complicated problem and will depend on how the algorithm is implemented.

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Deciding on an appropriate basic unit of computation can be a complicated problem and will depend on how the algorithm is implemented.

A good basic unit of computation for comparing the summation algorithms might be the number of assignment statements performed to compute the sum.

In the function `sumOfN()`:

```
long long sumOfN(long long n) {
    long long theSum = 0;
    for (long long i = 1; i <= n; i++) {
        theSum = theSum + i;
    }
    return theSum;
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}
```

The number of assignment statements is 1 (`the_sum = 0`) plus the value of n (the number of times we perform `the_sum = the_sum + 1`). We can denote this by a function, call it T, where $T(n) = n + 1$.

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```

The number of assignment statements is 1 (`the_sum = 0`) plus the value of n (the number of times we perform `the_sum = the_sum + 1`). We can denote this by a function, call it T, where $T(n) = n + 1$.

The parameter n is often referred to as the **size of the problem,** and we can read this as **$T(n)$ is the time it takes to solve a problem of size n, namely $n + 1$ steps.**

We can then say that the sum of the first 100,000 integers is a bigger instance of the summation problem than the sum of the first 1,000. Our goal then is to show how the algorithm's execution time (steps) changes with respect to the size of the problem.

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It turns out that the exact number of operations is not as important as determining the most dominant part of the function. In other words, as the problem gets larger, some portion of the function tends to overpower the rest.

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It turns out that the exact number of operations is not as important as determining the most dominant part of the function. In other words, as the problem gets larger, some portion of the function tends to overpower the rest.

The **order of the magnitude of the function** describes the part of $T(n)$ that increases the fastest as the value of n increases. Order of magnitude is often called Big O notation (for order) and written as $O(f(n))$.

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In the above example, $T(n) = n + 1$. As n gets larger, the constant 1 will become less and less significant to the final result. If we are looking for an approximation for $T(n)$, then we can drop the 1 and simply say that the running time is $O(n)$.

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It is important to note that the 1 is certainly significant for $T(n)$. However, as n gets large, our approximation will be just as accurate without it.

As another example, suppose that for some algorithm, the exact number of steps is . When is small, say 1 or 2, the constant 1005 seems to be the dominant part of the function. However, as gets larger, the term becomes the most important! .

As another example, suppose that for some algorithm, the exact number of steps is $T(n) = 5n^2 + 27n + 1005$. When n is small, say 1 or 2, the constant 1005 seems to be the dominant part of the function. However, as n gets larger, the n^2 term becomes the most important! .

In fact, when n is really large, the other two terms become insignificant for the final result. Again, to approximate $T(n)$ as n gets large, we can ignore the other terms and focus on $5n^2$.

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In fact, when n is really large, the other two terms become insignificant for the final result. Again, to approximate $T(n)$ as n gets large, we can ignore the other terms and focus on $5n^2$.

In addition, the coefficient 5 becomes insignificant as n gets large. We would say then that the function $T(n)$ has an order of magnitude $f(n) = n^2$, or simply that it is $O(n^2)$.

Sometimes the performance of an algorithm depends on the **exact values of the data rather than simply the size of the problem.** For these kinds of algorithms we need to characterize their performance in terms of best-case, worst-case, or average-case performance.

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The worst-case performance refers to a particular data set where the algorithm performs especially poorly, whereas a different data set for the exact same algorithm might have extraordinarily good (best-case) performance.

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The worst-case performance refers to a particular data set where the algorithm performs especially poorly, whereas a different data set for the exact same algorithm might have extraordinarily good (best-case) performance.

However, in most cases the algorithm performs somewhere in between these two extremes (average-case performance).

A number of very common order of magnitude functions will come up over and over as you study algorithms:

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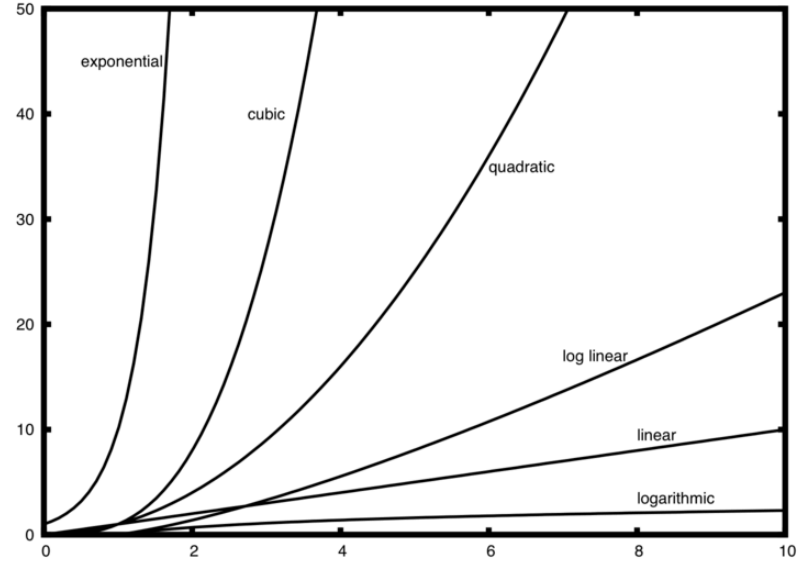
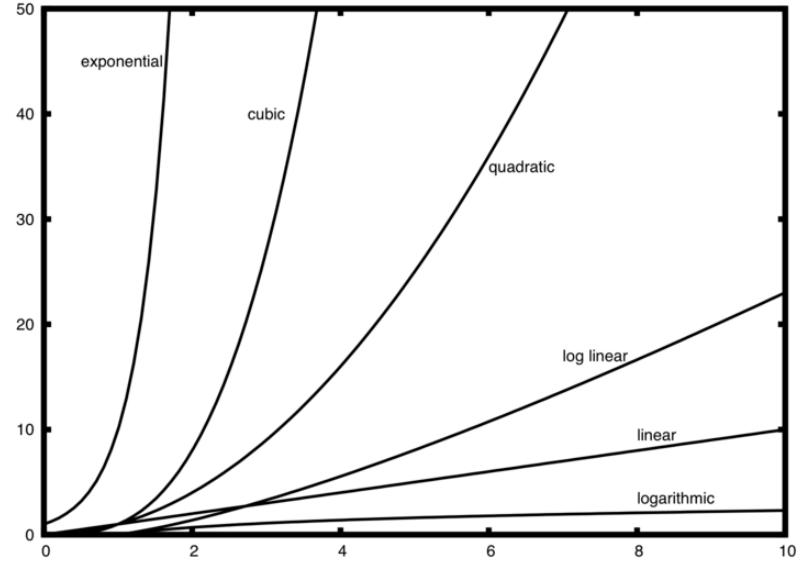
f(n)	Name
1	Constant
log(n)	Logarithmic
n	Linear
n log(n)	Log linear
n^2	Quadratic
n^3	Cubic
2^n	Exponential


50
40
30
20
10
0
0
2
4
6
8
10
exponential
cubic
quadratic
log linear
linear
logarithmic



Notice that when n is small, the functions are not very well defined with respect to one another. It is hard to tell which is dominant.

As a final example, suppose that we have the fragment of `C++` code:

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In [8]:
```
using namespace std;
int n = 100;
// Start of the code
int a = 5;
int b = 6;
int c = 10;
for (int i = 0; i < n; i++) {
    for (int j = 0; j < n; j++) {
        int x = i * i;
        int y = j * j;
        int z = i * j;
    }
}
for (int k = 0; k < n; k++) {
    int w = a * k + 45;
    int v = b * b;
}
int d = 33;
```

As a final example, suppose that we have the fragment of `C++` code:

In [8]:
```cpp
using namespace std;
int n = 100;
// Start of the code
int a = 5;
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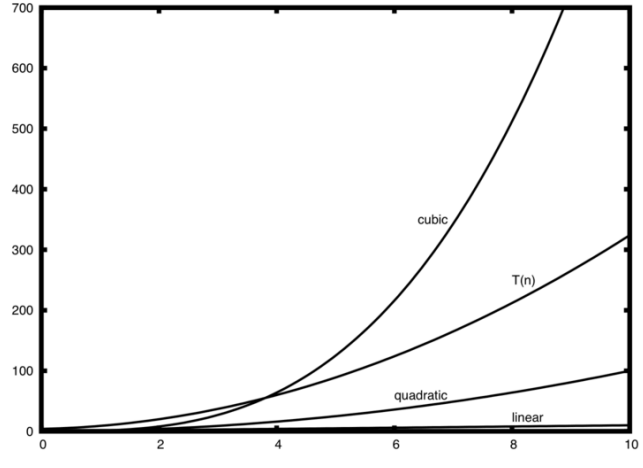
for (int i = 0; i < n; i++) {
    for (int j = 0; j < n; j++) {
        int x = i * i;
        int y = j * j;
        int z = i * j;
    }
}
for (int k = 0; k < n; k++) {
    int w = a * k + 45;
    int v = b * b;
}
int d = 33;
```

- The first part is the constant 3, representing the three assignment statements at the start of the fragment.
- The second part is $3n^2$ due to the nested iteration.
- The third part is $2n$ and the fourth part is the constant 1, representing the final assignment statement.

This gives us . We can see that the term will be dominant and therefore this code is . All of the other terms as well as the coefficient on the dominant term can be ignored as grows larger!

This gives us $T(n) = 3 + 3n^2 + 2n + 1 = 3n^2 + 2n + 4$. We can see that the n^2 term will be dominant and therefore this code is $O(n^2)$. All of the other terms as well as the coefficient on the dominant term can be ignored as n grows larger!



In [ ]:
```
display_quiz(path+"BigO1.json", max_width=800)
```

Given the following code fragment, what is its Big O running time?

```
int test = 0;
for (int i = 0; i < n; i++) {
    for (int j = 0; j < n; j++) {
        test = test + i * j;
    }
}
```

- ○ $O(n^2)$
- ○ $O(n)$
- ○ $O(\log n)$
- ○ $O(n^3)$

In [ ]:
```
display_quiz(path+"BigO2.json", max_width=800)
```

Given the following code fragment, what is its Big O running time?

```
int test = 0;
for (int i = 0; i < n; i++) {
    test = test + 1;
}
for (int j = 0; j < n; j++) {
    test = test - 1;
}
```

- ○ $O(n^3)$
- ○ $O(n^2)$
- ○ $O(\log n)$
- ○ $O(n)$

In [ ]:
```
display_quiz(path+"BigO3.json", max_width=800)
```

Given the following code fragment, what is its Big O running time?

```
int i = n;
while (i > 0) {
    int k = 2 + 2;
    i = i / 2;
}
```

- ○ $O(\log n)$
- ○ $O(n^3)$
- ○ $O(n)$
- ○ $O(n^2)$

2.4 An Anagram Detection Example

A good example problem for showing algorithms with different orders of magnitude is the classic anagram detection problem for strings. One string is an **anagram** of another if the second is simply a rearrangement of the first. For example, "heart" and "earth" are anagrams. The strings "python" and "typhon" are anagrams as well!

A good example problem for showing algorithms with different orders of magnitude is the classic anagram detection problem for strings. One string is an **anagram** of another if the second is simply a rearrangement of the first. For example, "heart" and "earth" are anagrams. The strings "python" and "typhon" are anagrams as well!

For the sake of simplicity, we will assume that the two strings in question are of **equal length** and that they are made up of symbols from the set of 26 lowercase alphabetic characters.

A good example problem for showing algorithms with different orders of magnitude is the classic anagram detection problem for strings. One string is an **anagram** of another if the second is simply a rearrangement of the first. For example, "heart" and "earth" are anagrams. The strings "python" and "typhon" are anagrams as well!

For the sake of simplicity, we will assume that the two strings in question are of **equal length** and that they are made up of symbols from the set of 26 lowercase alphabetic characters.

Our goal is to write a boolean function that will take two strings and return whether they are anagrams.

2.4.1 Solution 1: Anagram Detection Checking Off

Our first solution to the anagram problem will:

1. Check the lengths of the strings

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1. Check the lengths of the strings
2. Check to see that each character in the first string actually occurs in the second. If it is possible to check off each character, then the two strings must be anagrams.

Checking off a character will be accomplished by delete the element. However, since strings in `Python` are immutable, the first step will be to convert the second string to a `list`. Each character from the first string can be checked against the characters in the list and if found, checked off by delete it.

```
bool anagramSolution1(string s1, string s2) {
    bool stillOK = true;
    if (s1.length() != s2.length()) stillOK = false;   // Step 1
    vector<char> aList(s2.begin(), s2.end());
    unsigned pos1 = 0;
    while (pos1 < s1.length() && stillOK) {            // Step 2
        unsigned pos2 = 0;
        bool found = false;
        while (pos2 < aList.size() && !found) {
            if (s1[pos1] == aList[pos2]) found = true;
            else pos2 = pos2 + 1;
        }
        ...
```

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        unsigned pos2 = 0;
        bool found = false;
        while (pos2 < aList.size() && !found) {
            if (s1[pos1] == aList[pos2]) found = true;
            else pos2 = pos2 + 1;
        }
        ...
```

```
        ...
        if (found) aList.erase(aList.begin() + pos2);
        else stillOK = false;
        pos1 = pos1 + 1;
    }
    return stillOK;
}
```

(complete listing: `dscpp/anagram.hpp`)

```
bool anagramSolution1(string s1, string s2) {
    bool stillOK = true;
    if (s1.length() != s2.length()) stillOK = false;    // Step 1
    vector<char> aList(s2.begin(), s2.end());
    unsigned pos1 = 0;
    while (pos1 < s1.length() && stillOK) {             // Step 2
        unsigned pos2 = 0;
        bool found = false;
        while (pos2 < aList.size() && !found) {
            if (s1[pos1] == aList[pos2]) found = true;
            else pos2 = pos2 + 1;
        }
        ...
```

```
        ...
        if (found) aList.erase(aList.begin() + pos2);
        else stillOK = false;
        pos1 = pos1 + 1;
    }
    return stillOK;
}
```

(complete listing: `dscpp/anagram.hpp`)

In [9]:

```
#include <iostream>
#include "dscpp/anagram.hpp"
using namespace std;

int main() {
```

```
    cout << boolalpha << anagramSolution1("apple", "pleap") << endl;
    return 0;
}
```

true

Expected output: `true` — every character of `s1` was checked off in `s2`.

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Each of the n characters in `s1` will cause an iteration through up to n characters in the list from `s2`. Each of the n positions in the list will be visited once to match a character from s1. The number of visits then becomes the sum of the integers from 1 to n.
Therfore, $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$!

Expected output: `true` — every character of `s1` was checked off in `s2`.

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Therfore, $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$!

As n gets large, the n^2 term will dominate. Therefore, this solution is $O(n^2)$.

2.4.2 Solution 2: Sort and Compare

Another solution to the anagram problem will make use of the fact that even though `s1` and `s2` are different, they are anagrams only if they consist of exactly the same characters.

Another solution to the anagram problem will make use of the fact that even though `s1` and `s2` are different, they are anagrams only if they consist of exactly the same characters.

So if we begin by sorting each string alphabetically from a to z, we will end up with the same string if the original two strings are anagrams!

In [10]:
```cpp
#include <iostream>
#include <string>
#include <algorithm>
using namespace std;

bool anagramSolution2(string s1, string s2) {
    sort(s1.begin(), s1.end());
    sort(s2.begin(), s2.end());
    unsigned pos = 0;
    bool matches = true;
    while (pos < s1.length() && matches) {
        if (s1[pos] == s2[pos]) pos = pos + 1;
        else matches = false;
    }
    return matches;
}

int main() {
    cout << boolalpha << anagramSolution2("apple", "pleap") << endl;
    return 0;
}
```

```
true
```

In [10]:
```cpp
#include <iostream>
#include <string>
#include <algorithm>
using namespace std;

bool anagramSolution2(string s1, string s2) {
    sort(s1.begin(), s1.end());
    sort(s2.begin(), s2.end());
    unsigned pos = 0;
    bool matches = true;
    while (pos < s1.length() && matches) {
        if (s1[pos] == s2[pos]) pos = pos + 1;
        else matches = false;
    }
    return matches;
}

int main() {
    cout << boolalpha << anagramSolution2("apple", "pleap") << endl;
    return 0;
}
```

true

Expected output: `true` — after sorting, both strings become `aelpp`.

At first glance you may be tempted to think that this algorithm is $O(n)$, since there is one simple iteration to compare the n characters after the sorting process. However, the two calls to the `Python sort()` method are not without their own cost. As we will see in Chapter 5, sorting is typically either $O(n^2)$ or $O(n \log n)$, so the sorting operations dominate the iteration.

2.4.3 Solution 3: Brute Force

A brute force technique for solving a problem tries to exhaust all possibilities. For the anagram detection problem, we can simply generate a list of all possible strings using the characters from `s1` and then see if `s2` occurs.

A brute force technique for solving a problem tries to exhaust all possibilities. For the anagram detection problem, we can simply generate a list of all possible strings using the characters from `s1` and then see if `s2` occurs.

However, when generating all possible strings from `s1`, there are n possible first characters, $n-1$ possible characters for the second position, and so on. The total number of candidate strings is $n!$. Although some of the strings may be duplicates, the program cannot know this ahead of time!

It turns out that $n!$ grows even faster than 2^n as n gets large. In fact, if `s1` were 20 characters long, there would be $20! = 2,432,902,008,176,640,000$ possible candidate strings. If we processed one possibility every second, it would still take us $77,146,816,596$ years to go through the entire list!

2.4.4 Solution 4: Count and Compare

Our final solution to the anagram problem takes advantage of the fact that any two anagrams will have the same number of a's, the same number of b's, the same number of c's, and so on.

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In order to decide whether two strings are anagrams

1. First count the number of times each character occurs. Since there are 26 possible characters, we can use a list of 26 counters, one for each possible character. Each time we see a particular character, we will increment the counter at that position.

Our final solution to the anagram problem takes advantage of the fact that any two anagrams will have the same number of a's, the same number of b's, the same number of c's, and so on.

In order to decide whether two strings are anagrams

1. First count the number of times each character occurs. Since there are 26 possible characters, we can use a list of 26 counters, one for each possible character. Each time we see a particular character, we will increment the counter at that position.

2. In the end, if the two lists of counters are identical, the strings must be anagrams!

In [11]:

```
#include <iostream>
#include <string>
using namespace std;

bool anagramSolution4(string s1, string s2) {
    int c1[26] = {0};   // use ASCII code
    int c2[26] = {0};
    for (unsigned i = 0; i < s1.length(); i++) c1[s1[i] - 'a']++;
    for (unsigned i = 0; i < s2.length(); i++) c2[s2[i] - 'a']++;
    for (int i = 0; i < 26; i++) {
        if (c1[i] != c2[i]) return false;
    }
    return true;
}

int main() {
    cout << boolalpha << anagramSolution4("apple", "pleap") << endl;
    return 0;
}
```

```
true
```

In [11]:
```cpp
#include <iostream>
#include <string>
using namespace std;

bool anagramSolution4(string s1, string s2) {
    int c1[26] = {0};   // use ASCII code
    int c2[26] = {0};
    for (unsigned i = 0; i < s1.length(); i++) c1[s1[i] - 'a']++;
    for (unsigned i = 0; i < s2.length(); i++) c2[s2[i] - 'a']++;
    for (int i = 0; i < 26; i++) {
        if (c1[i] != c2[i]) return false;
    }
    return true;
}

int main() {
    cout << boolalpha << anagramSolution4("apple", "pleap") << endl;
    return 0;
}
```

true

Expected output: `true` — the two count arrays are identical.

Unlike the first solution, none of the loops are nested. The first two iterations used to count the characters are both based on . The third iteration, comparing the two lists of counts, always takes 26 steps since there are 26 possible characters in the strings. Adding it all up gives us steps. That is . We have found a linear order of magnitude algorithm for solving this problem!

Unlike the first solution, none of the loops are nested. The first two iterations used to count the characters are both based on n. The third iteration, comparing the two lists of counts, always takes 26 steps since there are 26 possible characters in the strings. Adding it all up gives us $T(n) = 2n + 26$ steps. That is $O(n)$. We have found a linear order of magnitude algorithm for solving this problem!

Before leaving this example, we need to say something about **space requirements**. Although the last solution was able to run in linear time, it could only do so by using additional storage to keep the two lists of character counts. **In other words, this algorithm sacrificed space in order to gain time.**

Unlike the first solution, none of the loops are nested. The first two iterations used to count the characters are both based on n. The third iteration, comparing the two lists of counts, always takes 26 steps since there are 26 possible characters in the strings. Adding it all up gives us $T(n) = 2n + 26$ steps. That is $O(n)$. We have found a linear order of magnitude algorithm for solving this problem!

Before leaving this example, we need to say something about **space requirements**. Although the last solution was able to run in linear time, it could only do so by using additional storage to keep the two lists of character counts. **In other words, this algorithm sacrificed space in order to gain time.**

On many occasions you will need to make decisions between time and space trade-offs. In this case, the amount of extra space is not significant. However, if the underlying alphabet had millions of characters, there would be more concern.

Exercise: Analyze the time complexity of the following code (You can consider as the power of 2 for approximation):

```
int i = 1;
while (i <= n) {
    for (int j = 1; j <= i; j++) {
        x += 1;
    }
    i *= 2;
}
```

Exercise: Analyze the time complexity of the following code (You can consider n as the power of 2 for approximation):

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        x += 1;
    }
    i *= 2;
}
```

Ans:

2.5~2.6 Performance of `C++` Data Structures: Vectors

The designers of `C++` had many choices to make when they implemented the `vector` data structure. To help them make the right choices they looked at the ways that people would most commonly use vectors, and they optimized the implementation so that the most common operations were very fast — in particular, a vector **doubles its capacity** when it runs out of room, making `push_back` amortized .

The designers of `C++` had many choices to make when they implemented the `vector` data structure. To help them make the right choices they looked at the ways that people would most commonly use vectors, and they optimized the implementation so that the most common operations were very fast — in particular, a vector **doubles its capacity** when it runs out of room, making `push_back` amortized $O(1)$.

Of course they also tried to make the less common operations fast, but when a trade-off had to be made the performance of a less common operation was often sacrificed in favor of the more common operation.

The designers of `C++` had many choices to make when they implemented the `vector` data structure. To help them make the right choices they looked at the ways that people would most commonly use vectors, and they optimized the implementation so that the most common operations were very fast — in particular, a vector **doubles its capacity** when it runs out of room, making `push_back` amortized $O(1)$.

Of course they also tried to make the less common operations fast, but when a trade-off had to be made the performance of a less common operation was often sacrificed in favor of the more common operation.

Two common operations are indexing and assigning to an index position. Both of these operations take the same amount of time no matter how large the list becomes. When an operation like this is independent of the size of the list, it is $O(1)$.

Another very common programming task is to grow a vector. `push_back()` is amortized , while inserting at the front with `insert(v.begin(), i)` is because every existing element must shift. Calling `reserve()` first avoids the reallocations entirely. Choosing the right tool makes your programs dramatically faster!

Another very common programming task is to grow a vector. `push_back()` is amortized $O(1)$, while inserting at the front with `insert(v.begin(), i)` is $O(n)$ because every existing element must shift. Calling `reserve()` first avoids the reallocations entirely. Choosing the right tool makes your programs dramatically faster!

Let's look at four different ways we might generate a list of first n integers starting with 0.

1. First we'll try a `for` loop and create the `list` by concatenation
2. We'll use `append()` rather than concatenation.

Another very common programming task is to grow a vector. `push_back()` is amortized $O(1)$, while inserting at the front with `insert(v.begin(), i)` is $O(n)$ because every existing element must shift. Calling `reserve()` first avoids the reallocations entirely. Choosing the right tool makes your programs dramatically faster!

Let's look at four different ways we might generate a list of first n integers starting with 0.

1. First we'll try a `for` loop and create the `list` by concatenation
2. We'll use `append()` rather than concatenation.

3. Next, we'll try creating the `list` using list comprehension
4. using the `range()` function wrapped by a call to the `list` constructor.

```cpp
void test1(int n) {            // insert at the front: O(n) per operation
    vector<int> v;
    for (int i = 0; i < n; i++) {
        v.insert(v.begin(), i);
    }
}

void test2(int n) {            // push_back: amortized O(1)
    vector<int> v;
    for (int i = 0; i < n; i++) {
        v.push_back(i);
    }
}
```

```cpp
void test1(int n) {            // insert at the front: O(n) per operation
    vector<int> v;
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void test2(int n) {            // push_back: amortized O(1)
    vector<int> v;
    for (int i = 0; i < n; i++) {
        v.push_back(i);
    }
}
```

```cpp
void test3(int n) {            // push_back with reserve: no reallocation
    vector<int> v;
    v.reserve(n);
    for (int i = 0; i < n; i++) {
        v.push_back(i);
    }
}

void test4(int n) {            // construct directly at the right size
    vector<int> v(n);
    for (int i = 0; i < n; i++) {
        v[i] = i;
    }
}
```

```cpp
void test1(int n) {            // insert at the front: O(n) per operation
    vector<int> v;
    for (int i = 0; i < n; i++) {
        v.insert(v.begin(), i);
    }
}

void test2(int n) {            // push_back: amortized O(1)
    vector<int> v;
    for (int i = 0; i < n; i++) {
        v.push_back(i);
    }
}
```

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void test3(int n) {            // push_back with reserve: no reallocation
    vector<int> v;
    v.reserve(n);
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        v.push_back(i);
    }
}

void test4(int n) {            // construct directly at the right size
    vector<int> v(n);
    for (int i = 0; i < n; i++) {
        v[i] = i;
    }
}
```

We will use `Python`'s `timeit` module. The module is designed to allow developers to make cross-platform timing measurements by running functions in a consistent environment and using timing mechanisms that are as similar as possible across operating systems!

To time these alternatives we use the `<chrono>` clock:

1. Record `steady_clock::now()` before and after the code you want to time
2. Convert the difference to seconds with `duration<double>`

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By default, `timeit` will try to run the statement one million times. When it's done it returns the time as a floating-point value representing the total number of seconds.

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1. Record `steady_clock::now()` before and after the code you want to time
2. Convert the difference to seconds with `duration<double>`

By default, `timeit` will try to run the statement one million times. When it's done it returns the time as a floating-point value representing the total number of seconds.

You can also pass `timeit` a named parameter called `number` that allows you to specify how many times the test statement is executed.

In [12]:

```
#include <iostream>
#include <vector>
#include "dscpp/dstimer.hpp"
using namespace std;

void test1(int n) { vector<int> v; for (int i = 0; i < n; i++) v.insert(v.begi
void test2(int n) { vector<int> v; for (int i = 0; i < n; i++) v.push_back(i);
void test3(int n) { vector<int> v; v.reserve(n);
                    for (int i = 0; i < n; i++) v.push_back(i); }
void test4(int n) { vector<int> v(n); for (int i = 0; i < n; i++) v[i] = i; }

int main() {
    void (*tests[])(int) = {test1, test2, test3, test4};
    const char* names[] = {"insert at front", "push_back", "with reserve", "di
    for (int k = 0; k < 4; k++) {
        DSTimer t;
        for (int r = 0; r < 1000; r++) tests[k](1000);
        printf("%-16s%9.2f ms\n", names[k], t.millis());
    }
}
```

insert at front 88.98 ms

push_back 4.41 ms

with reserve 3.44 ms

direct index 1.43 ms

In [12]:

```
#include <iostream>
#include <vector>
#include "dscpp/dstimer.hpp"
using namespace std;

void test1(int n) { vector<int> v; for (int i = 0; i < n; i++) v.insert(v.begi
void test2(int n) { vector<int> v; for (int i = 0; i < n; i++) v.push_back(i);
void test3(int n) { vector<int> v; v.reserve(n);
                    for (int i = 0; i < n; i++) v.push_back(i); }
void test4(int n) { vector<int> v(n); for (int i = 0; i < n; i++) v[i] = i; }

int main() {
    void (*tests[])(int) = {test1, test2, test3, test4};
    const char* names[] = {"insert at front", "push_back", "with reserve", "di
    for (int k = 0; k < 4; k++) {
        DSTimer t;
        for (int r = 0; r < 1000; r++) tests[k](1000);
        printf("%-16s%9.2f ms\n", names[k], t.millis());
    }
}
```

insert at front 88.98 ms

push_back 4.41 ms

with reserve 3.44 ms

direct index 1.43 ms

In the experiment above the statement that we are timing is the function call to `test1()`, `test2()`, and so on. You are probably very familiar with the `from... import` statement, but this is usually used at the beginning of a `Python` program file.

The helper `timeIt` runs each candidate 1000 times on a vector of 1000 elements, so tiny per-call overheads average out and the differences we see come from the algorithms themselves — the same reason Python's `timeit` runs statements many times in a clean namespace.

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From the experiment above it is clear that the `append()` operation is much faster than concatenation. It is interesting to note that the list comprehension is faster than a `for` loop with an `append()` operation.

You can look at the table below to see the Big O efficiency of all the basic vector operations.

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Operation	Big O Efficiency
index `[]`	O(1)
index assignment	O(1)
`append()`	O(1)
`pop()`	O(1)
`pop(i)`	O(n)
`insert(i, item)`	O(n)
`del` operator	O(n)
iteration	O(n)
contains (`in`)	O(n)
get slice `[x:y]`	O(k)
del slice	O(n)
set slice	O(n+k)
`reverse()`	O(n)
concatenate	O(k)
`sort()`	O(n log n)
multiply	O(nk)

You may be wondering about the two different times for removing an element. When `pop_back()` removes from the end of the vector it takes , but `erase(v.begin())` — removing the first element, or anywhere in the middle — is , because all elements after the removal point must shift left.

You may be wondering about the two different times for removing an element. When `pop_back()` removes from the end of the vector it takes $O(1)$, but `erase(v.begin())` — removing the first element, or anywhere in the middle — is $O(n)$, because all elements after the removal point must shift left.

In [13]:
```cpp
#include <iostream>
#include <vector>
#include "dscpp/dstimer.hpp"
using namespace std;

int main() {
    // time just the single erase/pop statement, 1000 times each
    vector<int> x(2000000);
    DSTimer t1;
    for (int r = 0; r < 1000; r++) x.erase(x.begin());
    printf("erase(begin()): %10.5f ms\n", t1.millis());

    vector<int> y(2000000);
    DSTimer t2;
    for (int r = 0; r < 1000; r++) y.pop_back();
    printf("pop_back():     %10.5f ms\n", t2.millis());
    return 0;
}
```

```
erase(begin()): 156.24149 ms

pop_back(): 0.00343 ms
```

The above shows one attempt to measure the difference between the two removals. Popping from the end is much faster than erasing from the beginning. However, this does not validate the claim that `erase(begin())` is while `pop_back()` is . To validate that claim we need to look at the performance of both calls over a range of vector sizes:

The above shows one attempt to measure the difference between the two removals. Popping from the end is much faster than erasing from the beginning. However, this does not validate the claim that `erase(begin())` is $O(n)$ while `pop_back()` is $O(1)$. To validate that claim we need to look at the performance of both calls over a range of vector sizes:

In [14]:
```
#include <iostream>
#include <vector>
#include "dscpp/dstimer.hpp"
using namespace std;

int main() {
    printf("%-10s%14s%12s\n", "n", "erase(begin)", "pop_back");
    for (int n = 2500000; n <= 10000000; n += 2500000) {
        vector<int> x(n);
        DSTimer te;
        for (int r = 0; r < 100; r++) x.erase(x.begin());
        double eraseT = te.millis();
        vector<int> y(n);
        DSTimer tp;
        for (int r = 0; r < 100; r++) y.pop_back();
        printf("%-10d%14.5f%12.5f\n", n, eraseT, tp.millis());
    }
}
```

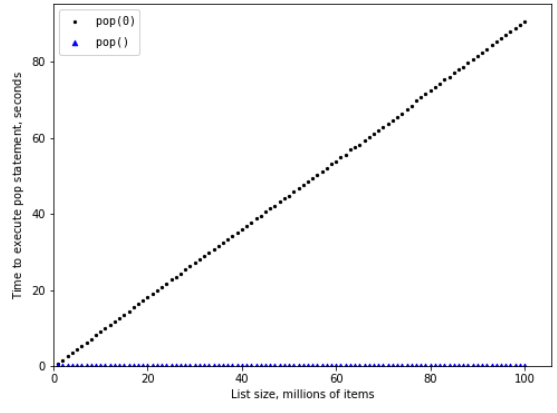
n erase(begin) pop_back

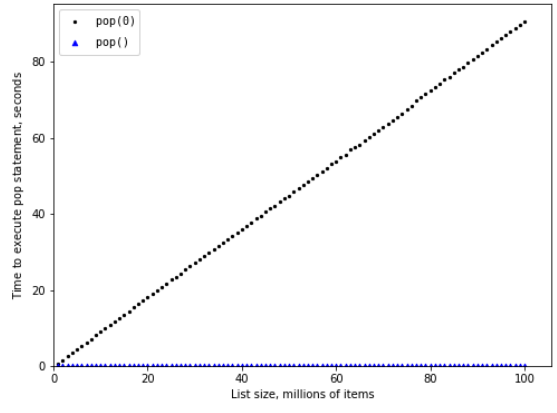
2500000 20.63682 0.00156

5000000 40.33524 0.00033

7500000 108.14922 0.00025

10000000 151.60363 0.00023


pop(0)
pop()
Time to execute pop statement, seconds
List size, millions of items
0
20
40
60
80
100



You can see that as the list gets longer and longer the time it takes to `pop(0)` also increases while the time for `pop()` stays very flat. This is exactly what we would expect to see for an $O(n)$ and $O(1)$ algorithm!

2.8 Hash Tables (unordered_map)

The second major `C++` associative structure is the **hash table**, `unordered_map`. As you probably recall, hash tables differ from vectors in that you can access items by a **key** rather than a position. Later in this course you will see that there are many ways to implement a hash table!

The second major `C++` associative structure is the **hash table**, `unordered_map`. As you probably recall, hash tables differ from vectors in that you can access items by a **key** rather than a position. Later in this course you will see that there are many ways to implement a hash table!

The thing that is most important to notice right now is that the get item and set item operations on a dictionary are $O(1)$. Another important dictionary operation is the `contains` operation. Checking to see whether a key is in the dictionary or not is also $O(1)$. The efficiency of all dictionary operations is summarized in Table below:

Operation	Big O Efficiency (average)
copy	O(n)
get item `[]`	O(1)
set item `[]`	O(1)
delete `erase`	O(1)
contains `count`	O(1)
iteration	O(n)

Operation	Big O Efficiency (average)
copy	O(n)
get item `[]`	O(1)
set item `[]`	O(1)
delete `erase`	O(1)
contains `count`	O(1)
iteration	O(n)

One important side note on dictionary performance is that the efficiencies we provide in the table are for **average-case performance** or **amortized analysis**. In some rare cases the contains, get item, and set item operations can degenerate into $O(n)$ performance, you can refer to Chapter 8 for more information.

For our last performance experiment we will compare the performance of the `contains` operation between vectors and hash tables. In the process we will confirm that searching a vector is while checking a key in an `unordered_map` is .

For our last performance experiment we will compare the performance of the `contains` operation between vectors and hash tables. In the process we will confirm that searching a vector is $O(n)$ while checking a key in an `unordered_map` is $O(1)$.

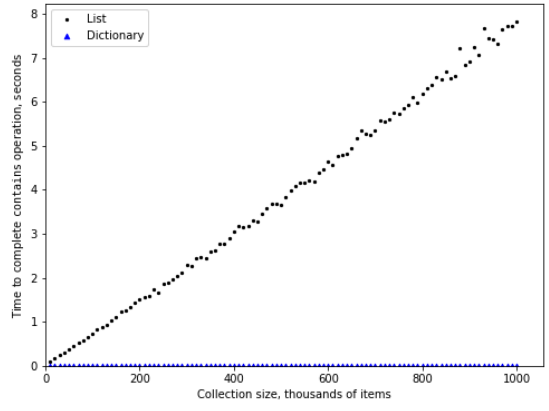
In [15]:
```
#include <unordered_map>
#include "dscpp/dstimer.hpp"
using namespace std;   // iostream/vector/algorithm preloaded by the kernel

int main() {
    printf("%-10s%10s%12s\n", "n", "vector", "hash table");
    for (int n : {250000, 500000, 1000000}) {
        vector<int> x(n);
        unordered_map<int, int> m;
        for (int j = 0; j < n; j++) { x[j] = j; m[j] = 0; }
        int hits = 0;
        DSTimer tv;
        for (int r = 0; r < 100; r++)
            hits += (find(x.begin(), x.end(), rand() % n) != x.end());
        double vecT = tv.millis();
        DSTimer tm;
        for (int r = 0; r < 100; r++) hits += m.count(rand() % n);
        printf("%-10d%10.3f%12.3f\n", n, vecT, tm.millis());
    }
}
```

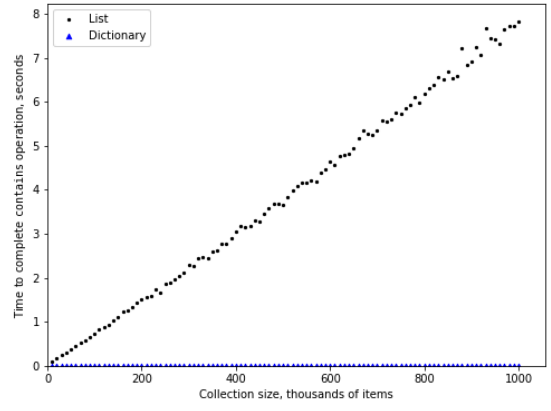
```
n vector hash table
```

250000 54.191 0.035

500000 108.132 0.036

1000000 176.195 0.043


List
Dictionary
Time to complete contains operation, seconds
Collection size, thousands of items
0
1
2
3
4
5
6
7
8
0
200
400
600
800
1000



You can see that the `dictionary` is consistently faster. You can also see that the time it takes for the `contains` operator on the `list` grows linearly with the size of the `list`. This verifies the assertion that the `contains` operator on a `list` is $O(n)$. It can also be seen that the time for the `contains` operator on a dictionary is constant even as the dictionary size grows.

In [ ]:
```
display_quiz(path+"dict1.json", max_width=800)
```

Which of the vector operations shown below is NOT $O(1)$?

- ○ vec.push_back(x)
- ○ vec[10]
- ○ vec.pop_back()
- ○ vec.erase(vec.begin())

In [ ]:
```
display_quiz(path+"dict2.json", max_width=800)
```

Which of the unordered_map operations shown below is $O(1)$ on average?

- ○ amap["x"] == 10
- ○ amap.count("x")
- ○ All are $O(1)$
- ○ amap["x"] = amap["x"] + 1
- ○ amap.erase("x")

Exercise: Devise an experiment to verify that the list index operator is .

Exercise: Devise an experiment to verify that the list index operator is $O(1)$.

In [16]:
```
// Your code here
```

In [ ]:

```
from jupytercards import display_flashcards
fpath = "flashcards/"
display_flashcards(fpath + 'ch2.json')
```

Next

>

References

1. Textbook: *Problem Solving with Algorithms and Data Structures using C++* (cppds), Chapter 2 — https://runestone.academy/ns/books/published/cppds/index.html